

## A. EVALUATING EXISTING RESEARCH

- Be skeptical but not cynical when reading published research. [Poor study design](#) and [selective reporting](#) are common features that, along with inadequate [statistical power](#) and [many forms of bias](#), decrease the reliability and replicability of much published research.
- Be wary of studies that (i) are published in [predatory journals](#); (ii) [emphasize statistical over biological significance, or P-values over effect sizes](#); (iii) report results from analyses that were unplanned or not reproducible; (iv) draw causal inferences from non-experimental research [without sufficient evidence](#); (v) [reference primarily supportive works while downplaying/ignoring contrary findings](#); or (vi) fail to discuss study limitations or alternative interpretations of their findings. These points should also inform B-D.
- [Distinguish exploratory \(hypothesis generating\) from confirmatory \(hypothesis testing\) research](#). Given its crucial role, exploratory research (ER) should be common and discoverable, but [research culture favours confirmatory research](#) (CR), causing much ER to be disguised as CR, raising the risks of [hindsight bias](#) and Hypothesizing After Results are Known (HARKing), and [decreasing research reliability](#).
- [Pilot studies](#) often blend ER with CR, and commonly [deviate from their intended use](#), reducing their reliability. Interpret them carefully.
- [P-values are most meaningful in pre-registered CR addressing plausible hypotheses, and least meaningful in ER disguised as CR](#).
- Consider conducting a [systematic review](#) of published research, including critically evaluating risk of bias following standardized [protocols](#).

## B. PLANNING YOUR RESEARCH

- [Consider research partnerships from the outset](#) Ideally, those who could benefit from or be impacted by the research would be directly involved through respectful collaboration.
- Complete ethics approvals and initiate a [data management plan](#) (DMP) prior to any data collection, ensuring ongoing respect for and adherence to community-specific engagement protocols, data sovereignty policies, and data management procedures, as per [FAIR](#) and [CARE](#) guidelines.
- Consider using [version-controlled reporting systems](#).
- Keep track of all contributors and their contributions from the outset; consider using [Dragon Kill Points](#), or the [MeRiT](#) or [CRedit](#) frameworks for reporting contributions to project outputs. Encourage all contributors to acquire an [ORCID](#) to facilitate discoverability and accurate attribution.
- Specify whether the research is ER or CR (or includes both), and provide rationale(s).
- Clearly define questions and/or hypotheses, and devise an appropriately designed observational or experimental study; [unless subjects/units are randomly assigned to treatments, the study is observational or quasi-experimental](#).
- Ethical and/or logistical constraints often necessitate observational or [quasi-experimental designs](#), in which case [causal inference is more challenging \(compared to experiments\) but possible](#), requiring [careful pre-planning, and meeting myriad criteria and assumptions](#).
- For experimental studies, ensure [proper controls and randomization, and implement blinding where possible](#).
- For all studies, define the target population, [scope of inference](#), dependent and independent variables, units of observation, and replication (sample sizes).
- Choose statistical methods best suited to addressing the question(s) / hypotheses; new methods are rarely needed.
- Multiple alternative data pre-processing and/or analysis methods may legitimately be suitable, so consider evaluating the robustness of the findings to these varied protocols (i.e. a form of [sensitivity analysis](#)).
- Plan to [accommodate non-Gaussian error distributions, multivariate responses](#) and/or [hierarchical/non-independent sampling designs](#) where appropriate.
- Specify [how model assumptions will be checked and violations dealt with](#).
- For CR, clearly define what would constitute a meaningful effect size, and [specify what will constitute evidence \(i\) consistent with and \(ii\) contrary to each hypothesis](#).
- For CR, consider [prospective power analyses](#) that employ the planned statistical methods and plausible effect sizes.
- Especially when power is limited, acknowledge uncertainty and [focus on sound study design and data useability](#).
- For CR, plan to [adjust P-values for multiple testing](#) when appropriate.
- For ER, justify sample sizes based on feasibility, resource constraints, or desired precision.
- Especially for CR, strongly consider [detailing your study design and analysis plan in a pre-registration](#) (PR), which can help improve research reliability. PRs shift effort appropriately towards the planning stage, where it has the greatest benefit. Deviations from PRs are [absolutely okay if transparently reported](#) (see section C).
- Have your research plan reviewed by both subject matter experts and others (possibly as a [registered report](#)).
- Foster a [peer community](#) that values providing and receiving constructive criticism.

## C. CONDUCTING YOUR RESEARCH

- Careful planning (part B) helps make the “conducting research” stage smoother.
- Document procedures transparently (ideally with [versioning](#)), detailing all deviations from planned protocols. Data quality control and updates to data management plans are ongoing responsibilities.

## D. REPORTING, INTERPRETING, AND COMMUNICATING YOUR RESEARCH FINDINGS

- Transparently report all evaluations of [model assumptions](#), and remedies to violations, as per analysis plans.
- [Prioritize reporting and interpreting effect sizes, their uncertainty \(e.g. confidence intervals\), and their real-world relevance](#).
- When appropriate, report effect sizes alongside *P*-values.
- Absence of evidence (i.e. a non-significant result) is not necessarily evidence of absence.
- Statistical significance does not necessarily imply biological relevance.
- Do not imply causation with words like “cause” or “affect” unless appropriately justified (e.g. from experimental results); consider whether “is associated with” is more suitable.
- Use effective visualizations: [reveal, don't conceal data](#).
- Clearly report sample sizes for all treatments, each analysis, and in each relevant figure and table.
- Maximize the [accessibility and inclusivity of outputs](#).
- There should be no surprise analyses: [each analysis should be linked to a pre-specified question/hypothesis](#).
- Adding unplanned analyses is sometimes warranted (e.g. as a follow-up to planned), but these tend to [yield inflated false positive rates](#) (e.g. due to [researcher degrees of freedom](#)), so label them clearly as unplanned; only planned analyses retain the intended statistical relevance.
- Interpret findings objectively and discuss alternative interpretations where appropriate.
- Avoid using [persuasive language](#), overselling, or over-generalizing your findings.
- Do not extrapolate findings beyond the target population without sufficient justification.
- Speculation can be valuable when labeled clearly as such.
- Evaluate your draft manuscript against points in section “A-1” above.

## E. ENHANCING THE REPLICABILITY, DISCOVERABILITY, ACCESSIBILITY, AND REACH OF YOUR RESEARCH

- Ensure [data](#) and [executable analysis code](#) are as open as possible and as closed as necessary according to [FAIR](#) and [CARE](#) guidelines, archived with appropriate metadata, and shared to facilitate independent replication.
- Publish Open Access and/or a [preprint](#), and follow [best practices to maximize discoverability when writing your title, abstract, and keywords](#).

*These are guidelines not requirements: every step taken towards more reliable, reproducible research is valuable and should be celebrated.*